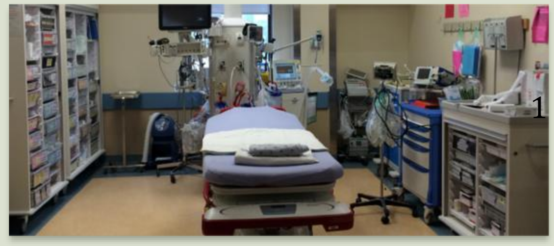
Anaphylaxis Case



Section I: Scenario Demographics

Scenario Title:	Anaphylaxis Case
Date of Development:	01/06/2015 (DD/MM/YYYY)
Target Learning Group:	☒ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

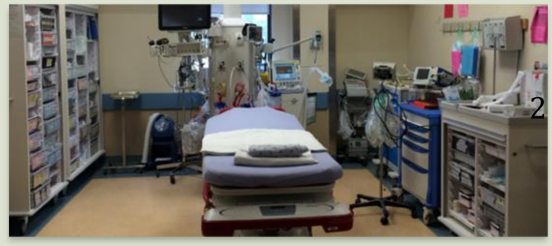
Scenario Developer(s):	Kyla Caners
Affiliations/Institution(s):	McMaster University
Contact E-mail (optional):	kcaners@gmail.com

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to common errors in the management of anaphylaxis
CRM Objectives:	1. Provides clear instructions to team members with closed-loop communication 2. Shares mental model so that all team members are aware of urgency of interventions and level of concern
Medical Objectives:	1. Gives repeated doses of epinephrine for anaphylaxis 2. Considers all possible adjuncts to epinephrine in the anaphylaxis patient 3. Recognizes impending potentially difficult airway

Case Summary: Brief Summary of Case Progression and Major Events
A 59-year-old male presents to the ED with anaphylaxis. He has already received a dose of epinephrine by EMS. On arrival, he will be wheezing and hypotensive with angioedema. Learners will be expected to provide repeat dosing of epinephrine as well as to start an epinephrine infusion in order for the patient to improve. They will also be expected to prepare for intubation. To highlight common errors in anaphylaxis treatment, a nurse will delay giving epinephrine unless specifically instructed to give it before other medications. The nurse will also attempt to give the cardiac epinephrine, requiring the team leader to clarify proper dosing. Once an epinephrine infusion has started, the patient's angioedema and breathing will improve.

Facilitators Required to Run Session
Instructors: 1-2 (one to observe and debrief, one to run mannequin) **Simple case, therefore could have one person do both roles
Confederate nurse: 1 (to delay epinephrine dosing and to attempt giving wrong epinephrine concentration)



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient	*Select most important dimension(s)*	☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

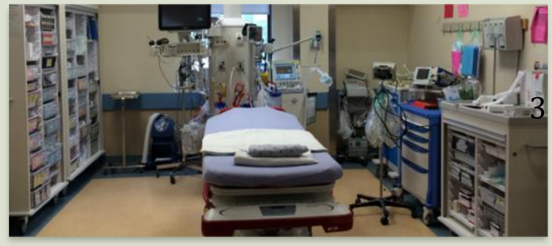
Confederates	Brief Description of Role
Nurse	Performs specific errors in medical management (delays giving epinephrine, potentially gives incorrect epinephrine dose)

B. Required Monitors		
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

C. Required Equipment		
☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

D. Moulage
Hives.

E. Approximate Timing					
Set-Up:	3 min	Scenario:	12 min	Debriefing:	20 min



Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

Report from EMS:
"This patient was recently prescribed Levofloxacin for a presumed pneumonia by his family MD. Approximately one hour after his first dose he developed a diffuse pruritic rash and felt acutely dyspneic. He denies any chest pain, syncope, fever or diaphoresis. He has not had Levofloxacin prior and there is no previous history of this."

"The highest SBP we could get was 90 by palp. Heart rate has been around 100. We've been unable to get an IV. Epi 0.5 IM x 1 has been given."

B. Patient Profile and History

Patient Name: George Smith	Age: 59	Weight: 70kg
Gender: ☒ M ☐ F	Code Status: Full	
Chief Complaint: allergic reaction		
History of Presenting Illness: as above		

Past Medical History:	HTN	Medications:	HCTZ 12.5mg PO daily
	Dyslipidemia		ASA 81mg PO daily
			Atorvastatin 40mg PO daily
Allergies: None			

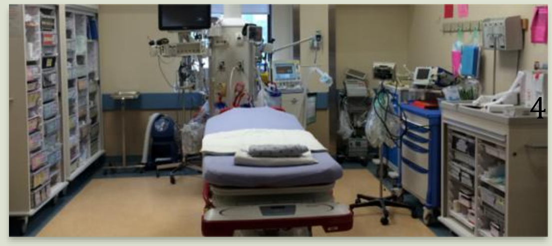
Review of Systems:	CNS:	No headache. Not dizzy. No syncope.		
	HEENT:	Nil.		
	CVS:	No chest pain.		
	RESP:	Very SOB since took antibiotic – about 30 min ago. No hx of same.		
	GI:	Nauseous. V/D. No abdominal pain.		
	GU:	Nil.		
	MSK:	Nil.	INT:	Rash

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display

HR: 105/min	BP: 90/50	RR: 24/min	O_2SAT: 91% RA
Rhythm: NSR	T: 36.9^{o}C	Glucose: 6.4mmol/L	GCS: 15 (E4 V5 M6)

General Status: Looks unwell.			
CNS:	Alert and able to answer questions		
HEENT:	**Angioedema to lips and IF THEY ASK:** **Also to posterior pharynx. No stridor. No acute airway obstruction.**		
CVS:	No murmur.		
RESP:	Diffuse expiratory wheeze.		
ABDO:	Soft, non-tender.		
GU:	Nil.		
MSK:	Nil.	SKIN:	**Diffuse hives**



Section VI: Scenario Progression

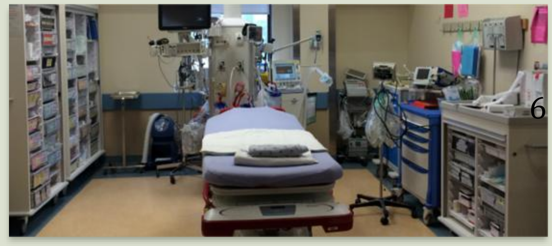
Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: NSR HR: 105/min BP: 90/50 RR: 24/min O_2SAT: 91% RA T: 36.6°C	Looks unwell. Angioedema and diffuse hives. Also wheeze. Increased WOB.	<u>Learner Actions</u> -☐ Apply O2, monitors -☐ Obtain 2 large bore IVs -☐ Bolus IV NS 1L -☐ Epi 0.5mg IM -☐ Ventolin neb 5mg (1ml) -☐ ±Epi 1:1000 5mg (5ml) neb -☐ Solumedrol 80-125mg IV -☐ Benadryl 50mg IV -☐ Ranitidine 50mg IV	<u>Modifiers</u> *Changes to patient condition based on learner action* - **RN to prompt** "Do you want Benadryl as well" after epi ordered → **If learners say yes, delay giving epi** - **RN to clarify epi dosing** (to have 1:10,000 cardiac epi in hand and ready to push iv) <u>Triggers</u> *For progression to next state* -5 min → **2. Stridor starting**
2. Stridor Starting BP → 80/40 HR → 115	Now has inspiratory stridor. Speaking 1 word sentences. Airway more edematous.	<u>Learner Actions</u> -☐ Repeat IV NS/RL bolus 1L -☐ Epinephrine infusion 0.1-0.5mcg/kg/min -☐ Epi 0.5mg IM (while prepare infusion) -☐ Prepare for intubation (difficult airway cart, call anesthesia) -☐ Epi 5mg (5ml) neb	<u>Modifiers</u> <u>Triggers</u> - Epi neb and epi infusion started → **3. Resolution** - 12 minutes → **3. Resolution**
3. Resolution BP → 100/60 HR → 90 O2 → 95%	**Patient states his breathing feels much better.**	<u>Learner Actions</u> -☐ Continue epi infusion -☐ Call ICU	**END CASE PRIOR TO INTUBATION**

Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results
None given during case.

Images (ECGs, CXRs, etc.)	
None required for case.	

Ultrasound Video Files (if applicable)	
None for case.	



Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video

Objectives	
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Sample Questions for Debriefing
1. What is the definition of anaphylaxis? Did this patient meet the criteria? 2. When is epinephrine given in anaphylaxis treatment? 3. What are indications for repeat epinephrine dosing? 4. What other medications are used in the treatment of anaphylaxis? 5. The nurse in this scenario almost gave the incorrect epinephrine dose. Do you have a sense of why epinephrine dosing is a common medication error? What are ways to prevent this? 6. How should medications be prioritized in the treatment of anaphylaxis?

Key Moments
Recognition of need for repeat epinephrine dosing
Clarification of epinephrine dosing and priority with nurse
Initiation of epinephrine infusion